

Are You Ready for the Singularity?



Over the past few million years, the world has been inhabited by at least ten different kinds of human animals. It's a nice line to consider: from the earliest version that first stood upright to the first to figure out fire, from the makers of stone tools and sharp weapons to the creators of complex ethical systems and inoffensive romantic comedies.

Naturally, those of us who inhabit the planet today like to imagine that we must be something special, what with our large and intricate brains, our flat-screen televisions and laser surgeries, our capacities for abstract art and space travel and self-awareness and so forth. We

like to believe that we are here and shall remain as the last kings of the planet, to be assured that we are the victorious culmination of evolution or the perfect triumph of a divine plan. To imagine that we are the best and final model of life on earth.

In October, a group of future-seekers came together in New York City to debate a curious idea: that perhaps we are not the final model. They gathered, more specifically, for the fourth annual Singularity Summit, a two-day conference about a concept that has been gathering momentum for the past few decades. The Singularity, in its simplest terms, is the moment when we make a machine (or become a machine) that is smarter than we are. But it's also everything that follows that moment: a chain of creation as that smart machine itself begins innovating, designing its own children, its own 2.0's and 3.0's and 4.0's, faster and faster, imagining new versions of themselves far better than we ever could, creating machines of a fantastic superintelligence, leading to an explosion of discovery and advancement unlike anything human history has ever seen. A kind of progress that will come so fast, and change the world so dramatically, that we can't even begin to imagine it.

Needless to say, this unimaginable state is a lot of fun to imagine. The 61-year-old futurist and inventor Ray Kurzweil, who serves as a director of the Summit's organizing body, the Singularity Institute for Artificial Intelligence (SI AI), is the best-known singer of the Singularity song. Onstage at the grand old theater inside the 92nd Street Y on Manhattan's Upper East Side, recapping the Summit's topics in a late-afternoon speech, Kurzweil seemed comfortable in his role as the movement's de facto leader, offering his perspective on the presentations, which had covered everything from near-term visions of radical life extension and eventual immortality to the prospect of building a brain from scratch. Kurzweil, though, was there not just to offer an executive summary. He also wanted to remind the audience of 800—mostly men, a mix of scholars and scientists and tech-heads, with a smattering of entrepreneurs, investors, and amateur enthusiasts mid in—that the predictions he has made about the

exponential growth of technology were proving to be on track, and that his road map to unimaginable progress was proceeding nicely. "What's really remarkable," he said, "is how predictable and inexorable a trajectory this is."

These predictions amount to something like a dream of happy apocalypse. But Kurzweil, a smallish, natty man who speaks in a flat, nasal tone, is a funny sort of prophet. Any cultish charisma or huckster's demagoguery is entirely absent from his presentation. He is simply there, he seems to say, to offer the evidence he has marshaled. In fact, he can come off as boring. The evidence he has marshaled, however, is not. In his view, it all points to the looming future moment when we humans become part machine, then fully machine, and then software—when we begin uploading the contents of our brains to computers, our minds and beings becoming nothing more than programs that exist on whatever hardware we export them to, with no need for a corporeal existence, no need for bodies that will eventually wrinkle, break down, and die. We'll live forever, Kurzweil says—formless, or in any form we choose.

And when will all this fun begin? Twenty generations from now? Fifty? A thousand? In *Radical Evolution*, the journalist Joel Garreau's 2005 book about new technologies and human enhancement, Garreau visits Michael Goldblatt, a former director of the Defense Sciences Office, a unit of DARPA (the Defense Advanced Research Projects Agency). Goldblatt shows Garreau two quotes from his files, from different sources but the same date: October 9, 1903. One, from an article in *The New York Times*, suggests that a flying machine "might be evolved by the combined and continuous efforts of mathematicians in from one million to ten million years." The other, from the diary of Orville Wright, says, "We started assembly today."

Cyborgs and conscious machines and immortal software beings are easily dismissed as science fiction—things that are unreal, things that none of us will ever experience. But many Singularitarians believe it could be just a few decades away—that we, now, are a generation that

will witness, that will *become*, the first generation of something new. That the next giant step of human evolution—or the end of human history altogether—is almost here.

Which is all very nice, and obviously entirely silly, and surely fertile terrain for Hollywood. Except that these ideas are not for screenplays, and the people behind them are not cranks. They are people with mainstream credibility, people with millions of dollars in grants from the federal government and even more in private funding, people with big jobs and big credentials, in academia and in business, from Stanford and MIT and Intel and IBM to start-ups you haven't heard of yet, but will. These are very smart men and women who have spent a lot of time doing a very creditable job at finding the right answers to whatever difficult questions they have pursued, and they have a lot of data that says they're right about this, no matter how strange it sounds. And what they believe, more or less, is this: Assembly has begun.

But what, exactly, are we assembling? "We're building God," Rick Schwall, a retired computer-system administrator who has attended the past three Singularity Summits, whispered to me in the theater at the Y. "And I, for one, think it's an excellent idea. We humans are terrible at being in charge. We're terrible to each other: We kill, cheat, torture, mutilate. So what we need is something else in charge—an intelligence programmed without these impulses. And the minute it gets to where it can talk, we ought to listen to it—and eventually turn the world over to it."

On the street outside the theater, I listened to two young men, taking a break from the lectures, idly batting around the implications of programming a superintelligence.

"If we're making God," said one—a skinny kid in jeans—"the question is, Which God?"

"What you're going to get at first is a bunch of weak gods running around," said the other.

"Demigods."

"Hercules, powered by Windows."

"Oh, dude, Hercules crashed again!"

"Then we get the big gods."

"Or the bad gods."

"But really: How do you design God?"

"Yeah—and how do you write a program that you know will make you the *right* God?"

"I'm not sure anyone's got the right language for that."

"I once tried to write a God with C++," the skinny kid said, "but it didn't take."

The God stuff, in fact, annoys most of the serious Singularity folk. They see it as a common dismissal of their vision, the idea that their research is nothing more than a cult dressed up as scholarship. Mitchell Kapor, the co-founder and former CEO of Lotus Development, has said of Singularity thinking, "It's intelligent design for the IQ-140 people." Others, like the virtual-reality pioneer Jaron Lanier, have compared Singularitarians to Christian evangelicals. The Singularitarians' response: Just because eternal life has been a cornerstone of most religions doesn't mean that a scientific vision of extreme evolution, of immortality, is a pseudo-spiritual pipe dream. We are working with data, they assert, not faith.

Inside the Y, in fact, most of the talks were focused not on dreamy visions of a radical future but on the nuts-and-bolts stepping-stones toward those dreams. Randal A. Koene, who directs the neuroengineering department at Tecnia, one of the largest private

research organizations in Europe, talked about his work with "whole-brain emulation"—an effort to fabricate a brain, essentially. Jürgen Schmidhuber, a professor of cognitive robotics at the Technical University of Munich, presented his research into an algorithm that would formalize human creativity and re-create it in machines. The science fiction writer and physicist Gregory Benford talked about his work in life-extension at Genescent, "the world's first company founded on the use of artificial biological selection," where, he said, their laboratory animals already live five times longer than their normal lifespan, with the human applications not too terribly far off.

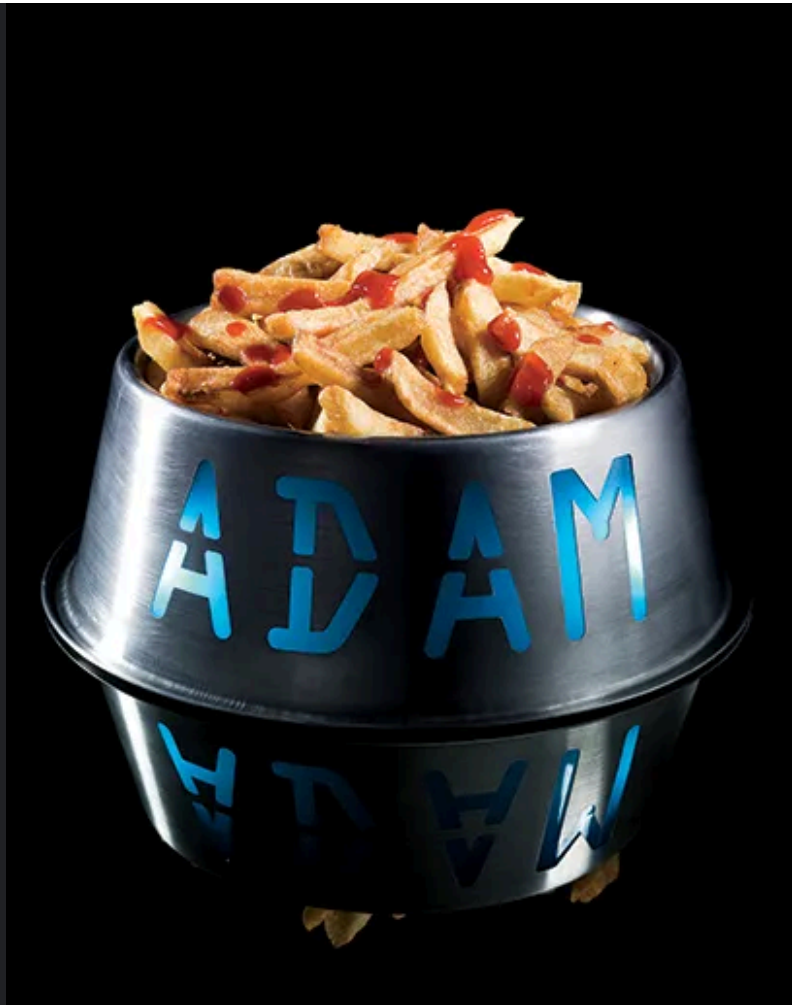
Vernor vinge, a 65-year-old mathematician, computer scientist, and science fiction writer, is usually credited with coining the term *the Singularity* in its tech-futurist sense. He first used it in a 1982 panel discussion and then, more famously, as the central concept of his 1993 essay "The Coming Technological Singularity: How to Survive in a Post-Human Era." Reading the implications of the steady curve upward in computer hardware, he predicted then that we would see superhuman intelligence by 2030.

A bald, owlish man with a wispy white beard, Vinge occasionally says something amusing enough to make himself giggle, an endearing sound, high and squeaky—a strangely *cute* sound to be coming from someone talking about whether machine intelligence will work with us or against us. At the previous Summit, the year before—held in Silicon Valley's unofficial capital, San Jose—he said to a few fans surrounding him, "The nearest analogue to the Singularity in the history of the earth is the rise of humans in the animal kingdom. Almost works, but not quite—that's not anywhere as dramatic a change as the Singularity. And yes, we know what happened when we came along, so this might not be good news for humans—unless we're *becoming* the next thing, not just being left behind by it."

"Look," I heard one guy say dismissively to a friend later, "sequencing the genome is all very nice and, yes, useful in the near term, fine, but

the fact of the matter is, we won't need it for very long, except as a sort of historical record for what we used to be."

The history—that is, us—may wish to express some skepticism about that prospect, but it's hard to deny that things are moving along in a strange and magical direction. We've already begun creating cyborgs, in case you've missed it: Doctors have introduced neural implants in the brains of paralysis patients—computer chips in the head—to restore brain and motor function. They have inserted telescopes in the eyes of the blind to give them sight. For amputees, they've created prosthetic hands that provide a real sense of touch. Consider for just a moment that the phone you hold in your hand gives you access to more information than any library has ever stored. How long before that phone gets integrated into your body? How long before we take that Bluetooth set in your ear, implant it in your brain, and give it wireless Internet, so that your brain becomes its own massively expanded and augmented search engine?



Meanwhile, there are AI (artificial intelligence) programs that have solved biology, physics, and math problems no human scholar has been able to crack. A consortium of European researchers is experimenting with a robot made to mimic the learning curve of a 3 1/2-year-old child, with the hope that the thing will eventually lead the way to a robot teenager. And in March 2007, when AI researchers in Switzerland set a group of programmed robots free in a controlled environment with "food" that sustained their batteries and "poison" that killed them, the robots learned to communicate, helping each other find the food and warning against eating the poison. Those robots that found the food survived, and their programming was lifted from their "bodies" and passed down to a new generation: survival of the fittest.

Not all Singularitarians agree, of course, on exactly how we'll reach the promised land. There are hotly debated theories of arriving via direct brain-computer interfaces, genetic engineering, or biological

enhancement. The most popular route, however, is still probably AGI—an artificial general intelligence, a machine that gets smart. But the big question remains: How do you create the software for a mind?

Ben Goertzel, who runs his own AI company, Novamente, and serves as the SIAI's director of research, is seeking the answer through an open-source project called OpenCog. Goertzel's notion, in brief, is to tap a global network of researchers to build an AGI not just by pieces and constituent programs but by creating a kind of overall mind-architecture. He's trying to create a baby machine intelligence that will eventually grow up. "Initially, an artificial-general-intelligence system may not do anything very useful," Goertzel said at the San Jose Summit. "I have three children. One of them is now 18, and he knows a lot now, but all three of them, when they were first born, were just completely useless. & If you didn't know what they were going to grow into, you'd just throw 'em in the garbage." The same is true, he was implying, of the early stages of a general machine intelligence. It *will* learn to learn, but you've got to give it time.

Goertzel, who favors jeans and T-shirts that say things like *nietzsche* is my co-pilot, is one of the cheerier optimists; he thinks he's only a few years away. He points out that "narrow AI"—AI designed for specific purposes—has been tremendously successful, and that the dream of AGI has mostly been left behind for these practical goals: an algorithm for predicting financial markets, for instance, or for mining data in your genome. But this, he says, is changing.

One of Goertzel's most visible projects is his work with virtual pets: programs that have been taught simple things and then set loose in digital worlds to learn. On a screen above the stage, he showed a virtual girl trying to teach a virtual dog to dance. "Essentially, the dog is supposed to be learning a program," he said. For a moment, the AI dog just watched her. "Now the dog's being an idiot," Goertzel said. "He's going to just sit there." At one point the dog walked over, considered his options, and rubbed his butt on the girl. "Well, now he's going to get kicked in the head," Goertzel said. Eventually, the dog

began to shake around jerkily, rising up on its hind legs and waving its paws: It had learned to dance, after a fashion.

I went to see Goertzel a few months later, partly because I'd been unable to get that virtual dog out of my head and partly because Goertzel came off as less interested in leading a movement and more like he was just having fun trying to do a really neat thing.

We met for lunch at a Chinese buffet near his home in suburban Maryland, half an hour from the center of Washington, D.C., where, among other pursuits, Novamente develops AI applications for government clients. ("They don't like us to talk about it," Goertzel said, managing to grimace and grin at the same time. "I can neither confirm nor deny that we have ever seen any data.")

I asked why the dream of making an AGI—a dream that has persisted for a half century with no appreciable success—was realistic now. What made him think he was special? "The main thing is that we live in a special time," Goertzel said. Advances in hardware, cognitive sciences, and problem-solving algorithms, he said, were beginning to constitute the necessary elements: "I feel the ingredients have now come together for the first time. Of course, the ingredients aren't enough. You still need the recipe." But Goertzel feels he knows, generally speaking, how to get at it. "Well, what can we do that narrow AI can't do?" he asked. "Put simply, it's contextualizing. We say a computer can play chess. But it can't, really. Not the way we do, anyway. It can't learn the rules of chess by you just telling it. It can't put the pieces back on the board if they get knocked over. It can't teach someone else. Because it doesn't understand chess in the broader context of life and the world. Children are also given limited contexts. But once they understand things in that context, you broaden the context for them, offer them wider worlds, and they learn and develop."

His plan calls for five phases: programming the basic components, followed by the creation of an artificial toddler, an artificial child, an artificial adult, and finally an artificial scientist. (He says his team is at

the start of phase two.) But what are the basic components? Human children are born with minds ready to do this; how do you program a mind to be ready to receive those wider worlds? "Take the dog," Goertzel said. "You could view it as a program whose job is to learn programs. The dog creates procedures and tries them out and analyzes the reaction to its trials. Then it synthesizes a new program which summarizes what it's learned from the previous tries and generates new programs to get better at it."

These virtual agents are some of the most evocative examples of this new research, because they actually do things. They don't demonstrate human-level intelligence (or even anything close to it), but they *are* entities bopping along on their own steam. Some of the most interesting work is being done in games. Michael Sellers, who runs a virtual-gaming company in Austin, Texas, called Online Alchemy, creates autonomous agents in a program that he calls the People Engine. At a cocktail hour celebrating the close of the San Jose Summit, he told me that each agent his company creates has its own emotions, motivations, memories, relationships, perceptions, and behaviors, and each learns from its own experiences as well as from other agents. But Sellers learned quickly how unpredictable these agents can be. "We taught them to eat, but we didn't teach them *what* to eat," Sellers said. "So they had a house, and they tried eating the house. That didn't work very well. So they tried to eat an apple tree; then they tried apples. But there just weren't enough apples, so they were still hungry. Meanwhile, they'd been programmed with an associative effect—so for instance, if they see a dog and they feel pain, they'll very quickly associate all dogs with pain. It's part of the way they learn. So anyway, this one [virtual] guy, Stan, he was around the apples while there were still apples, and because the apples made them happy, Stan made them happy, and when there were no more apples, it turned out they associated Stan with food. And so they tried to eat him, and that was a bad thing."

Yes, indeed: the bad thing. Let us not neglect the bad thing. There are a few bad things to consider, and they are very, very bad. The

moviemakers, of course, have named many of them for us already, from HAL and Colossus to the Terminator and the Matrix. Remember those robots that learned to avoid poison and find food? Over the succeeding generations, they began to compete, and the smartest eventually learned to protect their food supply by misleading other robots. That is, their own evolution led them to some of the most useful (and time-honored) human skills: lying, cheating, stealing, and murder.

So what happens when the tool gets smarter than the toolmaker? How long will it want to be a tool? Maybe *we'll* be its tools— or maybe it will experiment on us like we experiment on the lower animals. Or maybe we'll be its pets. Or maybe it will figure out very fast that the planet is better off without us, or that *it's* better off without us.

Some Singularitarians believe that if we can't predict what superintelligence will be like or what it will do, any hope of controlling it is futile. If it decides it's time for us to go, it's probably going to be time for us to go. In the view of the roboticist Hans Moravec, because "biological species almost never survive encounters with superior competitors," we can expect to be replaced by robots. Moravec thinks that's fine. "The robots will be a continuation of us, and they won't mean our extinction any more than a new generation of children spells the extinction of the previous generation of adults," he has said.

But it is not unreasonable to at least pause at this. Does being the aged parent of a hard drive sound like an ideal future? So much of what's best about the human species has come from error, from struggle, from pain. If you use superintelligent machines to solve all the problems poets and philosophers have been arguing about for millennia, you lose the need for poetry and philosophy. If you become a cyborg and then a computer program whose life experiences are all a virtual reality, what have you lost by shedding the meat of your body? Nothing but trouble, say the optimists. But trouble has its uses. (As does the meat of our bodies, obviously. Virtual sex and virtual steaks and virtual naps are going to have to get pretty damn good to

eclipse the real thing.) If everything is reduced to virtual perfection, what have we lost by losing our mistakes, our complicated confusions, our rough travels?

Bill Joy, the co-founder of Sun Microsystems, famously articulated some of these fears in his 2000 *Wired* essay, "Why the Future Doesn't Need Us." Joy wrote that a future with intelligent machines, self-replicating nanotechnology, and full genetic engineering is not only a vision of potentially losing something essential to humanity but also a world in which the chances that we could destroy ourselves are very, very real. Joy concluded that we should consider regulating these advances carefully. The Singularitarians don't necessarily disagree. Most of the community, though, would point out that turning away from invention is not a path human history has chosen much; we tend to innovate and then struggle to live with the results. If it can be invented, someone will invent it. The process *will* happen.

Not everyone agrees with that, either. Bill McKibben, in his 2003 book *Enough: Staying Human in an Engineered Age*, goes further than Joy. "We need to do an unlikely thing," he writes... "We need to survey the world we now inhabit and proclaim it good." He suggests that if we're given the ability to become as gods, we should choose to turn back from these advances before they overwhelm us. This, of course, is not a new attitude. Revolutionary technologies, throughout history, have typically been construed as offenses against a higher power, from the theft of fire to the cloning of sheep: our desire to play at the role reserved for the divine, or nature, or just the order of things that ought to be. The Singularity, however, is something quite different. It's not just an insult to God; it's taking his job.

Ray Kurzweil's book *The Singularity Is Near* is probably the most exhaustive work on the Singularity and, obviously, plays off the doomsayer's prediction of the end of times. Kurzweil, like Moravec, believes we don't particularly have to worry about the machines because we'll have become them. Kurzweil himself is well-known for his intention to live forever. He is attempting to extend his life through

diet and supplements, with the intention of hanging around long enough to reach the era of immortality. (He has also said that he wishes to bring his father back to life by making a genetic copy of him from DNA gathered by nanobots.)

Not surprisingly, his predictions are some of the loopyest sounding. The beginning of human-level AGI arrives around 2029; mind uploading comes soon after, along with cyborg bodies; eventually, most of us choose to get rid of our bodies altogether, opting to live inside software instead; and the full-on Singularity arrives in 2045 (a conservative estimate compared with Vinge's prediction of 2030, but five years sooner than that of Justin Rattner, Intel's CTO, who has put it at 2050). After that—well, it involves human machines thinking at the speed of light, followed by the conversion of all matter in the universe into massive computers. This continues as the vestiges of human individuality are transformed into an entirely communal, interconnected existence, which eventually evolves into a single giant perfect supercomputer-AI-human thinking being.

It's a vision worth a little scrutiny, even a little mockery. But it's also worth noting that Kurzweil has been right before, and often. Take some of the predictions from his 1990 book, *The Age of Intelligent Machines*, in which he called not only the explosion of the Internet but also the wireless revolution that came next. He's in the National Inventors Hall of Fame and has received the National Medal of Technology and Innovation, as well as a boatload of other awards and honorary degrees. When I went to see him at his lab in Massachusetts, he had just given a speech at the Army Science Conference, which had focused on "disruptive technology." His inventions range from optical character recognition to speech synthesis to computer programs that compose music. One of his earliest inventions, a machine that reads text to the blind, has now been perfected to the extent that it can transform a simple cell phone with a camera into a text reader. You just snap a picture of the page and it reads out loud to you.

Kurzweil's office is on the second floor of a nondescript building in suburban Boston. On the plasma screen by a reception desk, an AI program creates original paintings, erasing its work and starting anew every few minutes; on the bookshelf in his office, Kant's *Foundations of the Metaphysics of Morals* sits next to *The New Hacker's Dictionary*.

I asked him about the dismissal of Singularity theory as nothing more than a cultish faith in science fiction immortality, a so-called "rapture of the geeks" in which we all ascend to software heaven.

"So this is very often the form of criticism," Kurzweil said, tieless and looking a little tired, but still dapper in his dark, well-tailored suit, "that these ideas aren't real, that you're just adopting religious ideas. But my ideas originated from a scientific basis. A very practical one: I was trying to predict trends in order to time my own inventions. I started investigating technology trends because I realized that the reason inventions did not succeed was that inventors were getting the timing wrong. And I noticed that if you can measure the underlying information properties, it was amazingly predictable. So I began to examine that."

Kurzweil's fundamental argument for why the Singularity is nearer than you think is what he calls the Law of Accelerating Returns. People think of change as though it happens at a steady, consistent pace, he says, but in fact technological progress is accelerating exponentially; the *rate* of progress is doubling at regular intervals. That is, you start on a slow curve—2, 4, 8, 16, 32, 64—but pretty soon you've got a million, 2 million, 4 million, and you get into the billions and trillions without waiting too long. You can see this rate pattern, he likes to point out, in the race to sequence the genome. In 1990 biochemists were skeptical that we'd ever accomplish this, since at the outset they had completed 1/10,000th of it. They thought it would take at least a century. It was done by 2003. At this rate, Kurzweil says, the twenty-first century will see a thousand times more progress than the twentieth did.

"Barring some catastrophic meltdown, these future technologies will happen," he said, pausing for a sip of green tea. "But *how* has not been written yet."

To that end, in June 2009, Kurzweil and Peter Diamandis, CEO of the X Prize Foundation, opened the Singularity University, a nonprofit institution offering courses in subjects such as biotechnology, nanotechnology, and artificial intelligence. The school is backed by Google, among others, and housed at the NASA Ames Research Center in Moffett Field, California. Advisers, instructors, and trustees include academics and businesspeople ranging from Google co-founder Larry Page to George Smoot, the Berkeley professor who won the 2006 Nobel Prize in physics. The idea, on one level, is to bring together thinkers from various disciplines and let them work at the intersections of those fields, integrating and applying new technologies to the world's problems.

"There's a way that society can prepare constructively for this," Kurzweil said, "and we're not doing enough. This is a serious exercise." Not everyone finds it so serious. Peter Norvig, who is the head of research at Google and the co-author of *Artificial Intelligence: A Modern Approach*, the standard academic text on AI, told me, "I don't think there's a clear path there. You have most of the mainstream AI people saying, 'We're going about our business. Don't bother us with this millennialism.'" One prominent visionary in the world of information technology wrote to me, "As a matter of general policy, I don't give interviews on the Singularity for the same reason that astronomers don't give interviews on astrology."

"I can see why people would be resistant to this," Kurzweil said.

"Because I didn't do it in an afternoon, either. I've thought about this for decades. And it's taken me time to overcome my own incredulity, to really think through what this means: Can I personally accept becoming partly, or mostly, nonbiological? And I've come around to *Yeah, that'd be pretty cool.*"

This process of becoming something else would render many of our abstract philosophical puzzles suddenly concrete. Is being "human" strictly biological—just an evolutionary status as a member of the species *Homo sapiens*? Or is it a more theoretical construction, the species that attempts to transcend itself, that creates rules to prevent the abuse of its worst capacities, that drinks too much whiskey now and then? If all this happens—if, say, the computer program that is my consciousness is copied and uploaded and then copied again a hundred more times, a hundred backups in case I develop an error or my system crashes—are those copies "me," too? What if one me gets caught with a virtual prostitute and another runs for Congress and another wants to marry a toaster—do I get to complain? Can I just kill off the one running for Congress? What happens to society when we master the universe and learn there isn't a God—or when we find that there is?

In *The Singularity Is Near*, Kurzweil relates a conversation he had with Bill Gates, who says that he agrees with Kurzweil "99 percent" but asks whether we won't be giving up something by giving up human individuality. "That's just a limitation of biological intelligence," Kurzweil responds. "As humans, we try to merge with others also, but our ability to accomplish this is fleeting." Gates says, "Everything of value is fleeting," to which Kurzweil replies, "Yes, but it gets replaced by something of even greater value." And Gates agrees: "True, that's why we need to keep innovating."

But of course, what they're talking about is the end of anything being fleeting at all. They're talking about the end of disease, the end of mortality—the end of limits. Something with no boundaries, because this is where we live forever and become one with each other and the universe. This is a place and a time where there is no beginning of me and no end of you. A time wherein we are ourselves and everyone else, each of us everything, wherein we are the universe and the universe is us and I am you and you are me and if there is a walrus in the picture, or a toaster, or a robot, he's you and me, too. A time wherein we never die. Where nobody, and nothing, ever ends. This is

why the Singularity is so baffling, so frightening, so exciting: because what we know about life is that there are limits. And because this is the beyond. It's the end of ends, the end of the defining feature of all the life we know.

Then again, it's also the beginning of something. It's the beginning of absolutely everything we don't know.

[Dan Halpern](#) is a New York—based writer._